

LOGIC AND PHILOSOPHY — 2019 (Solved)

H.S. 2nd Year — Full Marks: 100 | Pass Marks: 30 | Time: Three Hours

Q.1 — Very Short Answer Type (1 mark each)

. (a) What is the main subject matter of Logic? [1]

Ans: The main subject matter of Logic is inference — that is, the laws and forms of valid reasoning by which we pass from known truths (premises) to new truths (conclusions).

. (b) Is the conclusion of Unscientific Induction certain? [1]

Ans: No, the conclusion of Unscientific Induction is not certain; it is only probable.

. (c) Is there 'inductive leap' in Perfect Induction? [1]

Ans: No, there is no inductive leap in Perfect Induction, because it examines every single instance of the class before drawing the conclusion.

. (d) The Paradox of Induction is connected with the Principle of the Uniformity of Nature. Is it true? [1]

Ans: Yes, this is true.

. (e) Which is the first stage of hypothesis suggested by Mill's definition of hypothesis? [1]

Ans: The first stage is Observation of facts.

. (f) What is the meaning of 'vera causa'? [1]

Ans: 'Vera causa' means a true or real cause — a cause whose existence is independently proved and not merely assumed for the sake of explanation.

. (g) What is the exact object of moral judgement? [1]

Ans: Voluntary action, together with the motive and intention behind it, is the exact object of moral judgement.

. (h) Name the theory which states that the external world can be directly known. [1]

Ans: Naive Realism (Direct Realism).

. (i) Is the conclusion of Method of Agreement certain? [1]

Ans: No, it is only probable, not certain.

. (j) Ethics is a positive science. Is it true? [1]

Ans: No, this is not true. Ethics is a normative science, not a positive science.

Q.2 – Q.13 — Short Answer Type (2 marks each)

2. State two points of differences between deduction and induction. [2]

Ans: (i) Deduction proceeds from general to particular, while Induction proceeds from particular to general. (ii) The conclusion of deduction is certain, whereas the conclusion of induction is only probable.

3. Why is the law of causation called the formal ground of induction? [2]

Ans: The law of causation is called the formal ground of induction because it is a self-evident, a priori, universal principle which is presupposed rather than proved by experience. It supplies the logical warrant or formal justification for passing from particular observed instances to a general conclusion, without itself being derived from those particular observations.

4. Define Scientific Induction. [2]

Ans: Scientific Induction is that form of induction in which a universal conclusion is drawn from the observation of particular instances by discovering the real causal connection between the antecedent and the consequent, on the basis of the Law of Causation and the Law of Uniformity of Nature.

5. State any two qualitative marks of causation. [2]

Ans: (i) Cause is the invariable and unconditional antecedent of the effect. (ii) Cause and effect are relative terms — the same phenomenon may be a cause in one relation and an effect in another.

OR

Define conjunction of causes with an example. [2]

Ans: When two or more causes combine together and act jointly to produce a single effect in which the contribution of each individual cause cannot be separately traced or distinguished, it is called conjunction of causes. Example: the mixing of several coloured lights to produce white light.

6. How many types of verification are there in a hypothesis and what are these? [2]

Ans: There are two types of verification of a hypothesis: (i) Direct Verification, in which the hypothesis is confirmed by direct observation or experiment, and (ii) Indirect Verification, in which the hypothesis is confirmed indirectly by showing that all the consequences deducible from it agree with observed facts.

OR

What do you mean by analogical hypothesis? [2]

Ans: An analogical hypothesis is a hypothesis formed on the basis of resemblance between a known case and an unknown case; because two things agree in certain known respects, it is supposed that they may also agree in some further unknown respect. Example: supposing that there may be life on another planet because it resembles the Earth in several respects.

7. Write two criticisms of Naive Realism. [2]

Ans: (i) Naive Realism cannot explain the occurrence of error or illusion in perception. (ii) It cannot explain why the same object appears differently to different perceivers or under different conditions (the relativity of perception).

8. Give a definition of Idealism in your own words. [2]

Ans: Idealism is that philosophical theory which holds that ultimate reality is essentially mental or spiritual in nature, and that the existence of the physical/external world depends upon, and cannot be conceived apart from, a perceiving or thinking mind.

9. Distinguish between fallacy of non-observation and fallacy of mal-observation. [2]

Ans: Fallacy of non-observation occurs when some fact or instance relevant to the inquiry is completely overlooked or missed during observation. Fallacy of mal-observation occurs when a fact is actually observed but is wrongly perceived or wrongly interpreted, so that its true nature is misrepresented.

10. Write two advantages of observation as the material ground of Induction. [1+1=2]

Ans: (i) Observation gives us direct, first-hand knowledge of facts exactly as they occur in nature, without artificial interference. (ii) It supplies the raw factual material on which the whole process of induction is built.

11. Why is the Idealism of Leibnitz called Pluralistic Idealism? [2]

Ans: Leibnitz held that ultimate reality does not consist of a single spiritual substance but of an infinite number of independent, simple, spiritual substances called Monads. Since he admits a plurality of ultimate spiritual reals instead of one, his idealism is called Pluralistic Idealism.

12. Define negative condition with the help of a suitable example. [1+1=2]

Ans: A negative condition is a condition whose absence (and not presence) is necessary for the occurrence of the effect. Example: the absence of dampness is a negative condition for a matchstick to catch fire when struck.

13. Write the name of one primary and one secondary quality of an object. [1+1=2]

Ans: Primary quality: extension (size/shape) of an object. Secondary quality: colour of an object.

Q.14 – Q.23 — Short/Descriptive Type (4 marks each)

14. State the qualitative marks of a cause. [4]

Ans: The qualitative marks of a cause are: (i) Cause is the unconditional antecedent of the effect — it does not depend on any other factor for its efficacy. (ii) Cause is the invariable antecedent of the effect — whenever the cause occurs, the effect must follow. (iii) Cause and effect are relative terms; the same event may be a cause with reference to one event and an effect with reference to another. (iv) Cause is qualitatively different in kind from its effect, though quantitatively equal to it (energy is neither created nor destroyed, only transformed).

OR

What do you mean by plurality of causes? Why is plurality of causes not scientifically acceptable? [2+2=4]

Ans: Plurality of causes is the doctrine which states that the same effect may, on different occasions, be produced by different causes. For example, death may be caused by drowning, poisoning, or disease. This doctrine is not scientifically acceptable because true science demands a strict, unconditional, one-to-one relation between a particular cause and a particular effect; when the apparently different causes of the 'same' effect are analysed carefully, it is found that the effects are not really identical in all respects, or that there is some common causal element underlying all the apparent causes. Hence the plurality is only apparent, not real.

15. Do you support Naive Realism? Give reasons in support of your view. [1+3=4]

Ans: No, I do not fully support Naive Realism. Although it is the commonsense view and appeals to ordinary experience, it suffers from serious defects: (i) it cannot explain the occurrence of error and illusion, since if perception is always a direct revelation of the object, illusory perception becomes inexplicable; (ii) it cannot account for the relativity of perception, i.e., why the same object appears different to different observers or under different conditions of light, distance, etc.; (iii) it fails to explain perceptual variations caused by physiological or psychological differences among perceivers (e.g., colour-blindness). For these reasons, Naive Realism, though simple and natural, cannot be accepted as a fully satisfactory theory of perception.

16. What is 'Crucial Instance'? Explain. [4]

Ans: A Crucial Instance is a particular instance or experiment which helps to decide between two or more rival hypotheses proposed to explain the same fact. When such an instance agrees with one hypothesis and disagrees with the others, it proves the former to be true and rejects the rest. The word 'crucial' comes from 'crux' (a cross), suggesting a point where two roads meet and one has to choose the correct path. For example, Foucault's pendulum experiment served as a crucial instance in favour of the hypothesis that the earth rotates on its axis, as against the rival hypothesis that it does not.

17. State two advantages and two disadvantages of the Method of Difference. [2+2=4]

Ans: Advantages: (i) It is essentially a method of experiment, and therefore its conclusion is highly certain. (ii) Only two instances (the instance of occurrence and the instance of non-occurrence, agreeing in every respect except one) are required to apply this method.

Disadvantages: (i) It cannot be applied where artificial experiment is not possible, e.g., in astronomy, sociology, etc. (ii) It cannot be applied to cases of permanent causes, since a permanent cause cannot be removed at will.

OR

Write two advantages and two disadvantages of the Method of Agreement. [2+2=4]

Ans: Advantages: (i) It is widely applicable, particularly to cases where artificial experiment is not possible (a method of observation). (ii) It can be applied even to permanent causes, unlike the Method of Difference. Disadvantages: (i) Its conclusion is only probable, not certain. (ii) It is liable to be vitiated by Plurality of Causes.

18. Briefly express any four characteristics of Scientific Realism. [4]

Ans: (i) Scientific Realism distinguishes between primary qualities (extension, figure, motion, number) and secondary qualities (colour, taste, smell, sound). (ii) It holds that primary qualities are objective and belong to the object itself, while secondary qualities are subjective and exist only in the perceiving mind. (iii) It accepts a representative or indirect theory of perception — we do not know the object directly but only through the ideas/representations it produces in the mind. (iv) Since perception is representative and not direct, Scientific Realism admits that error and illusion in perception are possible and explicable.

19. Name the philosopher who made the statement 'Esse Est Percipi'. Explain the meaning of the statement. [1+3=4]

Ans: The philosopher is Bishop George Berkeley. The Latin phrase 'Esse Est Percipi' means 'To be is to be perceived.' By this statement Berkeley meant that the existence of any physical object consists entirely in its being perceived by some mind; apart from being perceived (or perceiving, in the case of minds), nothing material exists independently. According to Berkeley there is no material substance underlying the qualities we perceive — objects are simply collections of ideas existing in the mind of the perceiver or, when not perceived by any finite mind, in the mind of God.

OR

Do you think that Hegel's idealism tries to reconcile idealism and realism? Explain. [4]

Ans: Yes, Hegel's Objective Idealism may be regarded as an attempt to reconcile idealism and realism. Hegel held that reality is one all-inclusive Absolute Spirit or Absolute Idea which realises and expresses itself through the entire process of nature and history. Since the Absolute Idea manifests itself concretely in the objective, external world of nature exactly as it is known to us, Hegel's system gives objective reality and independent status to the external world (satisfying realism) while still maintaining that the ultimate nature of that world is rational/spiritual (satisfying idealism). In this way, Hegel's idealism unites the realist's insistence on an objective world with the idealist's insistence on the spiritual/rational character of reality.

20. Explain briefly the Dharma and Artha as Purusarthas. [2+2=4]

Ans: Dharma: Dharma means righteous conduct or moral duty performed in accordance with the scriptures and one's station in life (varna and ashrama). It regulates and controls the pursuit of the other Purusarthas (Artha and Kama) and is regarded as the foundation of both worldly well-being and spiritual progress. Artha: Artha means wealth, material prosperity and the means of livelihood necessary for sustaining life in society. It provides the material basis without which neither Dharma (performance of duties) nor Kama (satisfaction of desires) can be properly pursued; hence it is regarded as an essential, though subordinate, end of human life.

21. Write a note on Plato's Idealism. [4]

Ans: Plato's Idealism, known as the Theory of Ideas or Forms, holds that the ultimate and truest reality consists of eternal, unchanging, universal Forms (Ideas) existing in a supersensible, non-spatio-temporal world. The physical objects we perceive through the senses are merely imperfect copies or shadows of these perfect Forms; for example, all beautiful things in the world merely 'participate in' or imitate the eternal Form of Beauty. Sense-experience gives only opinion (doxa), never true knowledge; true knowledge (episteme) is knowledge of the Forms, attained through reason alone. At the head of the hierarchy of Forms stands the Form of the Good, which is the ultimate source of all reality and knowledge. Thus Plato's Idealism is essentially Objective Idealism, since the Forms exist independently of any individual mind.

22. Write briefly the nature of religion. [4]

Ans: Religion is a system of belief and practice by which man relates himself to a Supreme or Divine Power that he regards as controlling the universe and his own destiny. Its nature involves three main elements: (i) a cognitive element — belief in the existence of a superhuman or divine reality; (ii) an emotional/affective element — feelings of awe, reverence, love, and dependence

towards that reality; and (iii) a conative element — a corresponding mode of conduct, worship, and way of life inspired by that belief. Religion thus unites knowledge, feeling, and will, and aims at man's spiritual welfare and communion with the Divine.

23. Briefly explain the four stages of hypothesis. [4]

Ans: According to Mill, a hypothesis passes through four stages: (i) Observation of facts — the collection of relevant facts which need explanation. (ii) Formulation of the hypothesis — a provisional guess or supposition suggested to explain the observed facts. (iii) Deduction of consequences — logically deriving further consequences that must follow if the hypothesis is true. (iv) Verification — testing these deduced consequences against actual facts by observation or experiment; if the consequences are confirmed, the hypothesis is accepted (or in some accounts elevated to a theory); if not, it is rejected or modified.

Q.24 – Q.27 — Long Answer Type (6 marks each)

24. State any three similarities and any three dissimilarities between Unscientific Induction and Analogy. [3+3=6]

Ans: Similarities: (i) Both are based on observed resemblance among instances rather than on a demonstrated causal connection. (ii) Both yield conclusions that are only probable, never certain. (iii) Neither method employs the Law of Causation directly to establish its conclusion.

Dissimilarities: (i) Unscientific Induction proceeds from particular instances to a general/universal conclusion, whereas Analogy proceeds from a particular instance to another particular instance. (ii) Unscientific Induction is based on mere repetition of instances without noting any specific point of resemblance, while Analogy is based on resemblance in certain definite, specified respects. (iii) The conclusion of Unscientific Induction is a universal proposition, while the conclusion of Analogy is a singular proposition about one particular case.

OR

Define Unscientific Induction and state its main characteristics. [1+5=6]

Ans: Unscientific Induction (Induction per Simple Enumeration) is that form of induction improperly so-called in which a universal conclusion is drawn merely from the observation of a large number of instances agreeing in a certain character, without any exception having been met with, and without discovering the real causal connection between the instances observed. Main characteristics: (i) it is based on simple enumeration of agreeing instances; (ii) no causal connection between antecedent and consequent is established; (iii) the conclusion, though stated as universal, is only probable and never certain; (iv) it does not employ the Law of Causation or methods of elimination; (v) the probability of the conclusion increases with the

number of instances observed and decreases with a single exception; (vi) it often paves the way for, and may later be corrected into, Scientific Induction.

25. Define the Joint Method of Agreement and Difference with a concrete example. State two advantages and two disadvantages of this method. [2+2+2=6]

Ans: Definition: The Joint Method of Agreement and Difference states that if two or more instances in which the phenomenon occurs have only one circumstance in common (while differing in all other respects), and two or more instances in which the phenomenon does not occur have nothing in common except the absence of that circumstance, then the circumstance in which alone the two sets of instances differ is the cause (or effect, or an indispensable part of the cause) of the phenomenon.

Example: Consider two groups of malaria patients similar in every other respect. One group is given quinine and recovers (positive/agreement instances all have quinine in common); the other group is not given quinine and does not recover (negative/difference instances agree only in the absence of quinine). Hence quinine is concluded to be the cause of recovery from malaria.

Advantages: (i) It is more reliable than the simple Method of Agreement, since it partly guards against the fallacy of Plurality of Causes. (ii) It can be applied in cases where direct experiment (Method of Difference) is not possible, being largely a method of observation.

Disadvantages: (i) It does not give a fully certain conclusion like the Method of Difference; the conclusion remains only highly probable. (ii) It is still liable, to some extent, to the fallacy of Plurality of Causes and Intermixture of Effects.

OR

Explain the Method of Residues with examples. [6]

Ans: The Method of Residues states that when a known complex effect is found to be produced by a complex cause, and we subtract from the total effect that part which is already known (by previous inductions) to be the effect of certain antecedents, the residue of the effect is the effect of the remaining antecedents. Symbolically, if ABC is followed by abc, and it is known that A is the cause of a and B is the cause of b, then C must be the cause of the residual effect c.

Example: The discovery of the planet Neptune is a classic example. Astronomers observed irregularities (perturbations) in the orbit of the planet Uranus that could not be fully explained by the gravitational pull of the then-known planets. After subtracting the known causes of the perturbations, a residual disturbance remained. This residue was attributed to the gravitational pull of an unknown planet, whose position was mathematically calculated in advance and the planet — later named Neptune — was subsequently discovered exactly where predicted.

26. Explain Aristotle's view on causation. [6]

Ans: According to Aristotle, to fully know and explain any object or event, we must know its cause, and he distinguished four kinds of causes: (i) Material Cause — the matter or substance out of which a thing is made; e.g., marble is the material cause of a marble statue. (ii) Formal Cause — the form, essence, plan or pattern which shapes the matter into a particular thing; e.g., the design or shape given to the marble is the formal cause of the statue. (iii) Efficient Cause — the agent or force that actually brings about the change, i.e., that which produces the effect by acting upon the matter; e.g., the sculptor who carves the statue is its efficient cause. (iv) Final Cause — the end or purpose for the sake of which the thing exists or the change takes place; e.g., the purpose of worship, decoration, or commemoration for which the statue is made is its final cause.

Of these four, Aristotle regarded the Final Cause as the most important, since he believed that everything in nature moves and develops towards a purpose or end (teleology). Modern logicians, however, mainly use the term 'cause' in the sense corresponding to Aristotle's Efficient Cause, i.e., the immediate active agent that produces the effect.

OR

Explain the Law of Causation as one of the formal grounds of Induction. [6]

Ans: The Law of Causation states that every event has a cause, and that the same cause, under the same conditions, will always produce the same effect. This law is regarded as a formal ground of induction because it is a universal, self-evident, a priori principle that is presupposed by, rather than derived from, any particular act of induction; without assuming this law in advance, no inductive generalisation from observed instances to unobserved instances would be logically justified. It thus supplies the formal or logical warrant that entitles us to pass from 'this cause has produced this effect in observed cases' to 'this cause will always produce this effect,' making it, along with the Law of Uniformity of Nature, one of the two formal (as opposed to material) grounds of induction.

27. 'Religion and Morality are mutually dependable' — Discuss it. [6]

Ans: Religion and Morality, though distinct, are closely related and mutually dependent aspects of human life. On the one hand, morality needs religion for its completion and support: religion provides the sanction and motive-power for moral conduct by relating moral duties to the will of God or to a Divine/Absolute purpose, and it gives moral life a deeper meaning, hope, and consolation (especially regarding the ultimate triumph of good over evil) which morality alone cannot supply. On the other hand, religion needs morality for its content and expression: a

religion devoid of moral conduct degenerates into mere ritual or superstition, and the value of any religion is largely judged by the quality of moral life it produces in its followers. Thinkers such as Kant held that morality leads to religion (the postulates of God, freedom, and immortality are required to make the moral life fully rational), while others such as Matthew Arnold held that 'religion is morality touched with emotion,' making morality the very substance of religion. Thus, although religion and morality can be conceptually distinguished — religion centering on man's relation to the Divine and morality on right conduct among men — in actual human life they support, enrich, and depend upon each other, and a completely satisfactory account of either requires reference to the other.

OR

What do you mean by Religion? Which is the most acceptable definition of Religion? Explain two characteristics of Religion. [2+2+2=6]

Ans: Religion may be understood as a system of belief in, and worship of, a superhuman or Divine Power that is regarded as governing the universe and human destiny, together with the way of life and conduct inspired by that belief. Among the various definitions offered (by Kant, Hegel, Matthew Arnold, Flint, and others), the definition given by Flint — that religion is man's belief in a being or beings, superhuman, who exercise a supervision and control over the affairs of the world, together with the feelings and conduct which flow from such belief — is generally regarded as the most comprehensive and satisfactory, since it covers the cognitive (belief), emotional (feeling), and conative (conduct) elements of religion without being confined to any one particular creed. Two characteristics of Religion: (i) Belief in a Supreme/Divine Power controlling the universe — a cognitive element common to all religions. (ii) A definite mode of conduct or way of life, including worship and moral practice, arising out of that belief — a conative/practical element that translates religious faith into everyday living.